

LAB EXPERIMENT RECORD

SET 4: PRODUCT INVENTORY MANAGEMENT

Department	AIDS
Course Code / Course Name	DSA0602 / Data Handling and Visualization
Experiment	Data Visualization using R and Tableau
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Date	1 October 2026
Day	Thursday

Aim

To visualize product inventory data using a bar chart, a stacked bar chart and a scatter plot in R, and to build an interactive dashboard using Tableau.

Dataset: Product Inventory

Product ID	Product Name	Quantity Available
1	Product A	250
2	Product B	175
3	Product C	300
4	Product D	200
5	Product E	220

Additional columns used: The given table contains only Product ID, Product Name and Quantity Available. For Programs 2 to 4, a Product Category and a Product Price column were added to the dataset:

Product ID	Product Name	Product Category	Product Price (\$)
1	Product A	Electronics	20
2	Product B	Electronics	35
3	Product C	Accessories	15
4	Product D	Accessories	25
5	Product E	Home Appliances	22

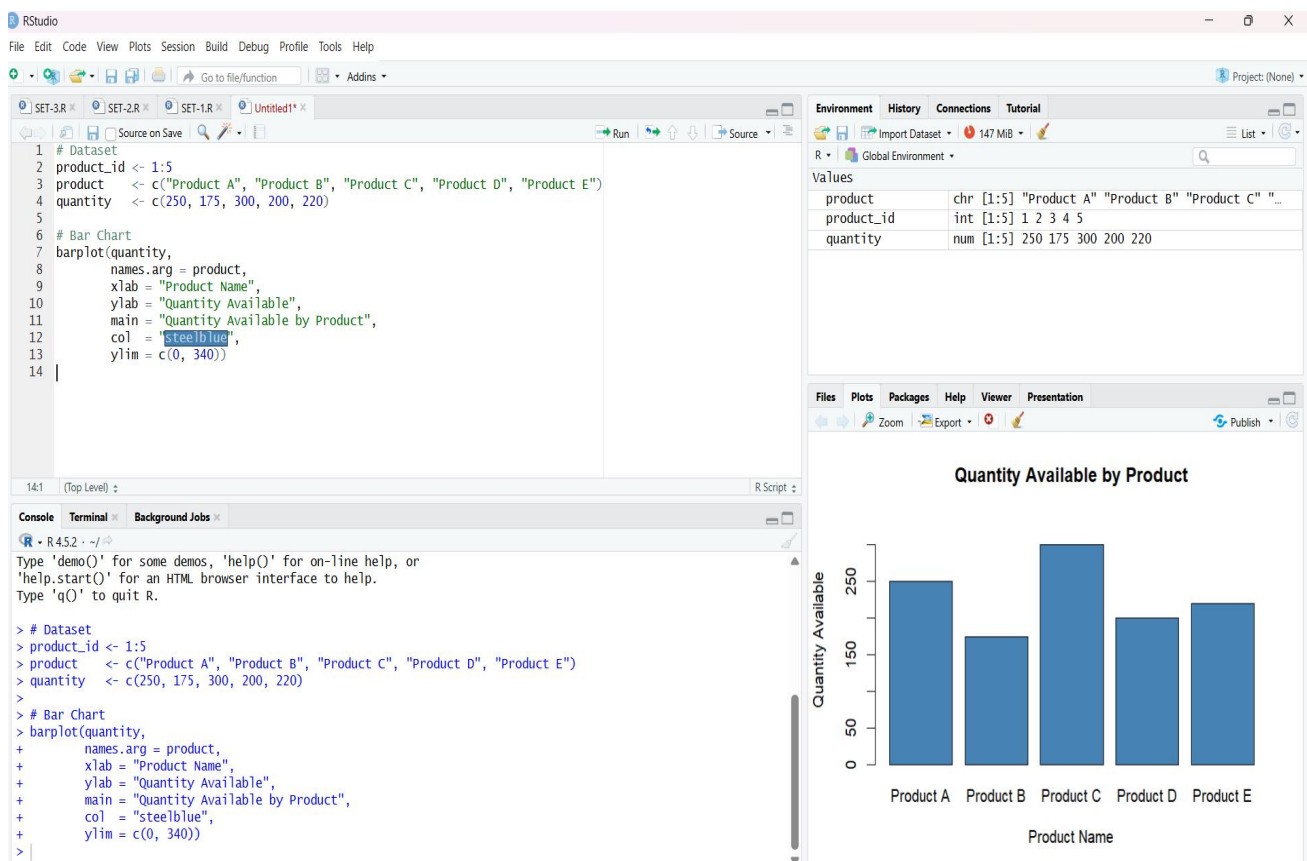
PROGRAM 1: Bar Chart of Quantity Available

Create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.

```
# Dataset
product_id <- 1:5
product    <- c("Product A", "Product B", "Product C", "Product D", "Product E")
quantity   <- c(250, 175, 300, 200, 220)

# Bar Chart
barplot(quantity,
          names.arg = product,
          xlab = "Product Name",
          ylab = "Quantity Available",
          main = "Quantity Available by Product",
          col = "steelblue",
          ylim = c(0, 340))
```

OUTPUT:



Output 1: Quantity Available by Product (Bar Chart)

Observation: Product C has the highest quantity available (300 units), followed by Product A (250). Product B has the lowest stock (175 units). The total inventory across all five products is 1,145 units, an average of about 229 units per product.

PROGRAM 2: Stacked Bar Chart by Product Category

Generate a stacked bar chart to show the quantity of each product within different product categories.

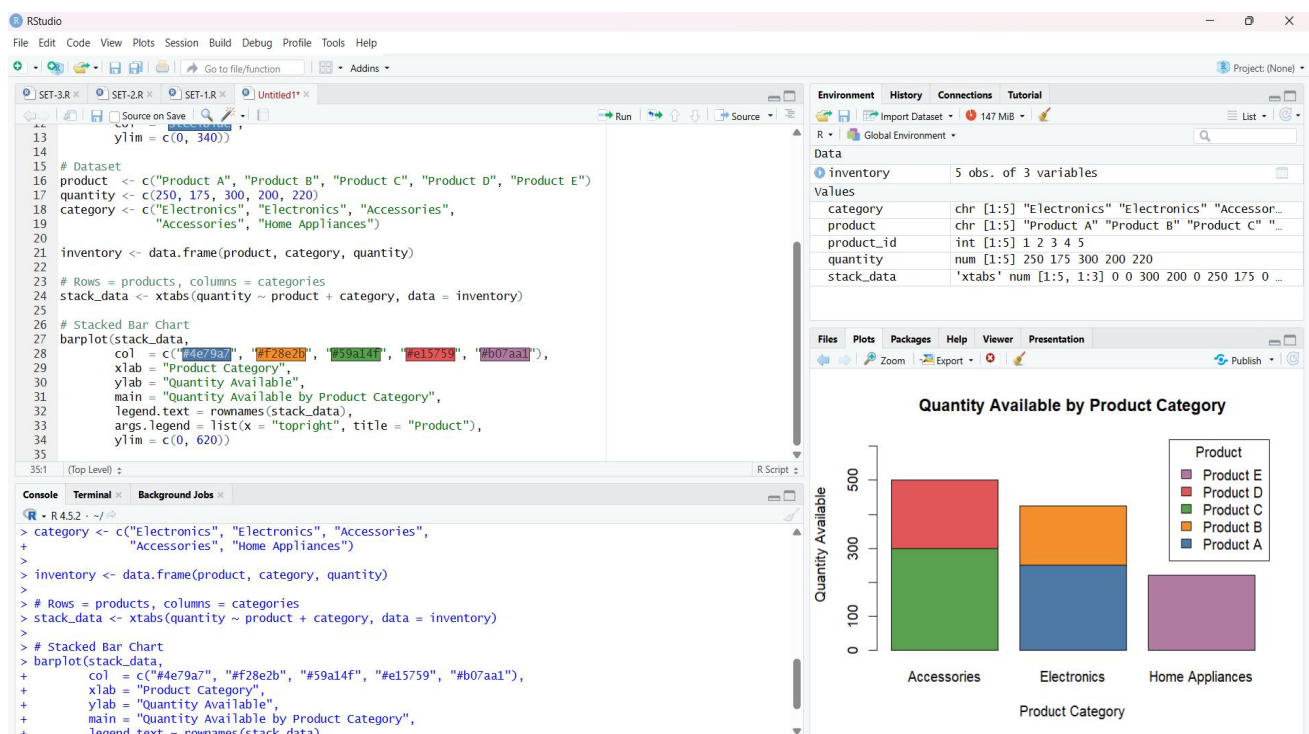
```
# Dataset
product <- c("Product A", "Product B", "Product C", "Product D", "Product E")
quantity <- c(250, 175, 300, 200, 220)
category <- c("Electronics", "Electronics", "Accessories",
             "Accessories", "Home Appliances")

inventory <- data.frame(product, category, quantity)

# Rows = products, columns = categories
stack_data <- xtabs(quantity ~ product + category, data = inventory)

# Stacked Bar Chart
barplot(stack_data,
        col = c("#4e79a7", "#f28e2b", "#59a14f", "#e15759", "#b07aa1"),
        xlab = "Product Category",
        ylab = "Quantity Available",
        main = "Quantity Available by Product Category",
        legend.text = rownames(stack_data),
        args.legend = list(x = "topright", title = "Product"),
        ylim = c(0, 620))
```

OUTPUT:



Output 2: Quantity by Product Category (Stacked Bar Chart)

Observation: Accessories has the largest total stock (500 units: Product C 300 + Product D 200), followed by Electronics (425 units: Product A 250 + Product B 175) and Home Appliances (220 units, Product E only). Each segment of a bar shows how much a single product contributes to its category.

PROGRAM 3: Scatter Plot of Price vs Quantity Available

Build a scatter plot to explore the relationship between product price and quantity available. Explain the findings.

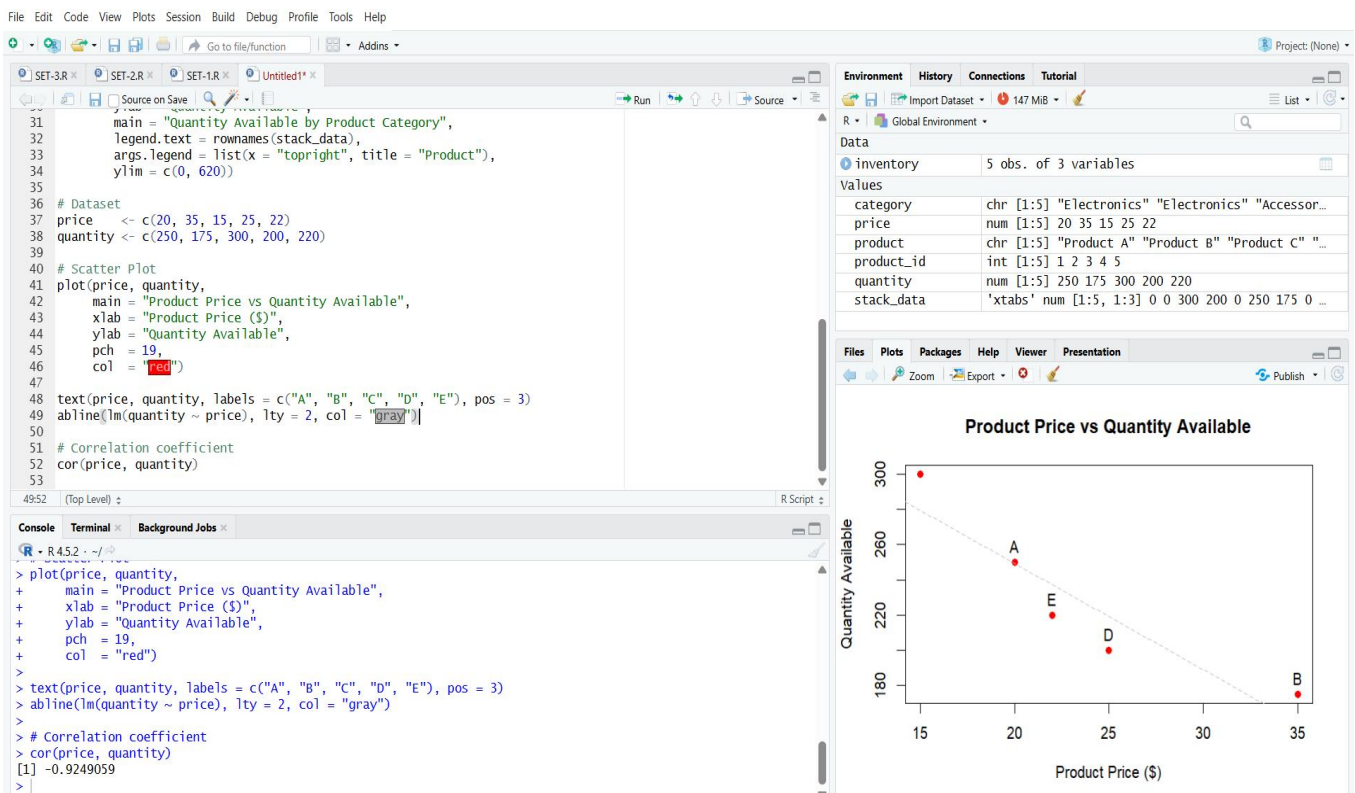
```
# Dataset
price    <- c(20, 35, 15, 25, 22)
quantity <- c(250, 175, 300, 200, 220)

# Scatter Plot
plot(price, quantity,
     main = "Product Price vs Quantity Available",
     xlab = "Product Price ($)",
     ylab = "Quantity Available",
     pch = 19,
     col = "red")

text(price, quantity, labels = c("A", "B", "C", "D", "E"), pos = 3)
abline(lm(quantity ~ price), lty = 2, col = "gray")

# Correlation coefficient
cor(price, quantity)
```

OUTPUT:



Output 3: Product Price vs Quantity Available (Scatter Plot)

Insight: The points follow a downward trend: as the product price increases, the quantity available decreases. Product C has the lowest price (\$15) and the highest quantity (300), while Product B has the highest price (\$35) and the lowest quantity (175). The correlation coefficient is about -0.92, which indicates a strong negative relationship. This suggests that lower-priced products are stocked in larger quantities than higher-priced products.

Note: The relationship is based on only five products, so it shows a pattern in this dataset and does not prove that price causes the change in quantity.

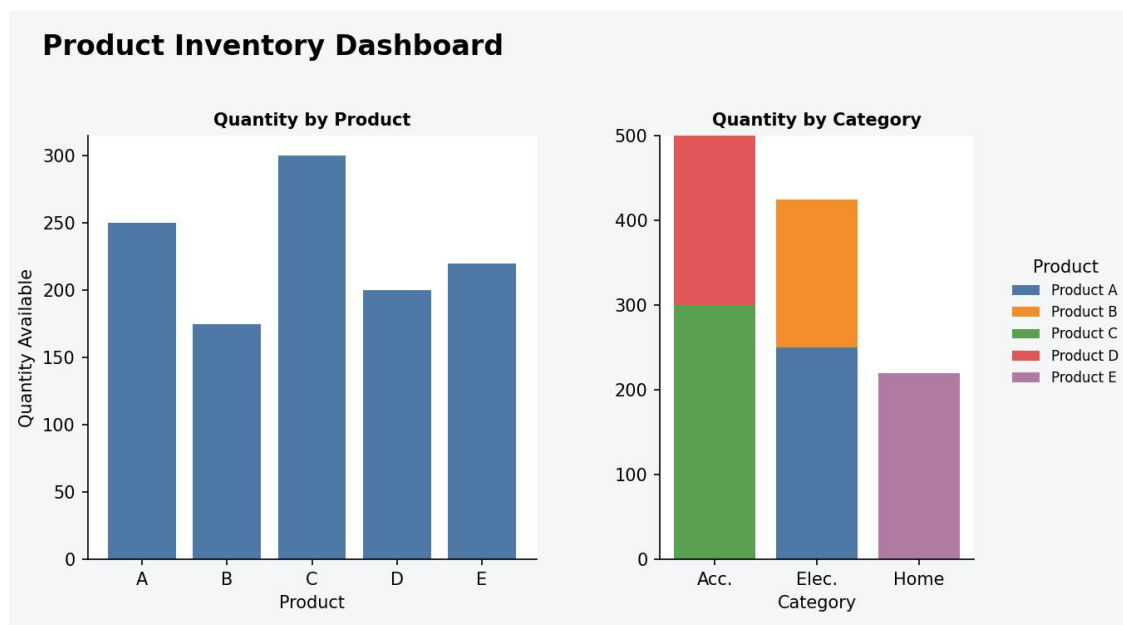
PROGRAM 4: Interactive Inventory Dashboard in Tableau

Develop a Tableau dashboard with the bar chart and stacked bar chart to allow users to interact with the data.

Steps in Tableau

1. Save the product inventory data (Product ID, Product Name, Product Category, Product Price, Quantity Available) as an Excel or CSV file. Open Tableau and connect to this file using Connect > Microsoft Excel / Text file.
2. Sheet 1 (Bar Chart): Drag Product Name to Columns and Quantity Available to Rows (SUM). Add Quantity Available to Label to show the values. Rename the sheet "Quantity by Product" and set the title and axis labels.
3. Sheet 2 (Stacked Bar Chart): Drag Product Category to Columns and Quantity Available to Rows (SUM). Drag Product Name to Color, so each bar is stacked by product. Rename the sheet "Quantity by Category".
4. Create a new Dashboard (Dashboard > New Dashboard). Set the size to Automatic, then drag both sheets onto the dashboard area.
5. Add interactivity: select each chart and click Use as Filter, so clicking a bar in one chart filters the other chart.
6. Add a Product Category filter and a Product Name filter (Show Filter > Apply to Worksheets > Selected Worksheets) so users can choose the data to display.
7. Add a dashboard title, "Product Inventory Dashboard", and a color legend. Check the dashboard in Presentation Mode to test the filters.

OUTPUT:



Output 4: Interactive Product Inventory Dashboard (Tableau)

Observation: The dashboard places the bar chart and the stacked bar chart side by side. Selecting a product bar or a category in either chart filters the other chart, and the filter controls let users view the inventory of one category or one product at a time.

Procedure

1. Define the product, category, price and quantity vectors in R.
2. Use the `barplot()` function to create a bar chart of quantity by product, with axis labels and a title.
3. Build a product-by-category table using `xtabs()` and pass it to `barplot()` to create the stacked bar chart.
4. Create equal-length vectors for price and quantity, then use `plot()` to generate the scatter plot and `cor()` to measure the relationship.
5. Load the inventory file into Tableau and build the bar chart and stacked bar chart worksheets.
6. Combine both worksheets in a Tableau dashboard and add filters and actions for interactivity.

RESULT:

The product inventory data was successfully visualized using a bar chart, a stacked bar chart and a scatter plot in R. An interactive Tableau dashboard was also created, combining the bar chart and stacked bar chart so that users can filter and explore the inventory data.

Key Functions and Tools Used

Function / Tool	Purpose
<code>barplot()</code>	Creates bar charts and stacked bar charts.
<code>xtabs()</code>	Builds the product-by-category table used for stacking.
<code>plot()</code>	Creates the scatter plot.
<code>abline()</code> / <code>lm()</code>	Adds a trend line to the scatter plot.
<code>cor()</code>	Calculates the correlation between price and quantity.
Tableau Worksheet	Builds the bar chart and stacked bar chart.
Tableau Dashboard	Combines worksheets with filters and actions.